

Analysis and Compact Modeling of Fast Detrapping From Bandgap-Engineered Tunneling Oxide in 3-D NAND Flash Memories

Minsoo Kim^{ID} and Hyungcheol Shin, *Senior Member, IEEE*

Abstract—We present the comprehensive analysis and the compact modeling methodology of fast electron detrapping from the bandgap engineered tunneling oxide (BE-TOX) in 3-D NAND flash memories. To this end, we develop a physics-based detrapping model considering the detailed electron dynamics in the BE-TOX and various emission paths of the detrapping mechanism. By the aid of the carrier density rates from the accurately calibrated model, we find that the major path of the charge loss can be direct tunneling (DT) or trap-to-band tunneling (TBT) depending on the temperature and the nitride layer position in the BE-TOX. In addition, we compare the model result and the fitted stretched exponential function. From those comparisons, we correlate several indicators (time constant and curve shape) of the retention characteristics with the major paths of the charge loss. Finally, our model facilitates the quantitative separation of the detrapping mechanism into different paths.

Index Terms—3-D NAND flash memories, bandgap engineering, compact modeling, detrapping mechanism, short-term retention.

I. INTRODUCTION

EVEN though 3-D charge-trap NAND flash memories are gradually replacing 2-D floating gate NAND flash memories [1], several issues on the data retention have continuously emerged [2], [3]. For the analysis of retention characteristics, many previous studies have attempted to separate the charge loss mechanisms [4]–[7]. Among the mechanisms, electron detrapping is unique as it occurs at the tunneling oxide (TOX) while others mainly originate from the charge trap nitride (CTN). Even though the TOX layer is not used for the purpose of the storage layer, the electrons are unexpectedly trapped in this layer during the memory operations and then detrapp over

time, creating a retention issue. After the bandgap engineering technology is adopted to improve the memory window [8], [9], the electron detrapping from the bandgap engineered (BE)-TOX has been reported to be more severe and even occur within very short time (~ 1 s) [6], [7] due to the shallow trap [10]. The state-of-the-art 3-D NAND flash memories have been adopting triple-level-cell (TLC) and quadruple-level-cell (QLC) technologies with much tighter threshold voltage (V_{th}) distributions. Since the detrapping from the BE-TOX deteriorates with program/erase (P/E) cycles [7] and shrinks read margins, this mechanism should be studied for the reliability enhancement in 3-D NAND flash. Although a lot of previous studies have suggested models on the charge loss from the CTN [11]–[15], the compact modeling of the electron detrapping from the BE-TOX has not been focused on.

Thus in this article, we propose a physics-based compact model of the fast electron detrapping mechanism during short-term retention. Since the detrapping mechanism is one of the fastest among the charge loss mechanisms introduced in the short-term retention of 3-D NAND flash and is not significantly affected by the cell patterns [6], [7], the model can be developed first and independent of other charge loss mechanisms. Starting from the Poisson equations of BE-TOX/CTN/blocking oxide (BOX) layers in cylindrical coordinates, we obtain the necessary electrostatic solutions for the model. By means of the previous charge loss models [11]–[15] and the 3-D technology computer-aided design (TCAD) simulation [16] including the physics in the BE-TOX trap, we develop a compact detrapping model at the BE-TOX which accurately reproduces all the possible paths of electrons and the consequent threshold voltage shift (ΔV_{th}). We separate the emission paths of the electrons into trap-to-band tunneling (TBT), direct tunneling (DT), and thermal emission (TE); and most of the charge loss is caused by TBT and DT. Contrary to the reported detrapping mechanism from the conventional TOX [17] which is mainly attributed to TE, TE is found to be almost negligible for BE-TOX during the short-term retention. The nitride layer in the BE-TOX far from the channel gives rise to the delayed detrapping characteristics while the temperature dependence and the total ΔV_{th} increase. In that regard, we suggest an appropriate guideline for designing the BE-TOX.

Manuscript received January 28, 2021; revised March 21, 2021 and April 21, 2021; accepted April 28, 2021. Date of publication May 14, 2021; date of current version June 23, 2021. The review of this article was arranged by Editor S. Alam. (Corresponding author: Minsoo Kim.)

The authors are with the Inter-University Semiconductor Research Center, School of Electrical Engineering and Computer Science, Seoul National University, Seoul 151-742, South Korea (e-mail: dhn2213@snu.ac.kr).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TED.2021.3077202>.

Digital Object Identifier 10.1109/TED.2021.3077202

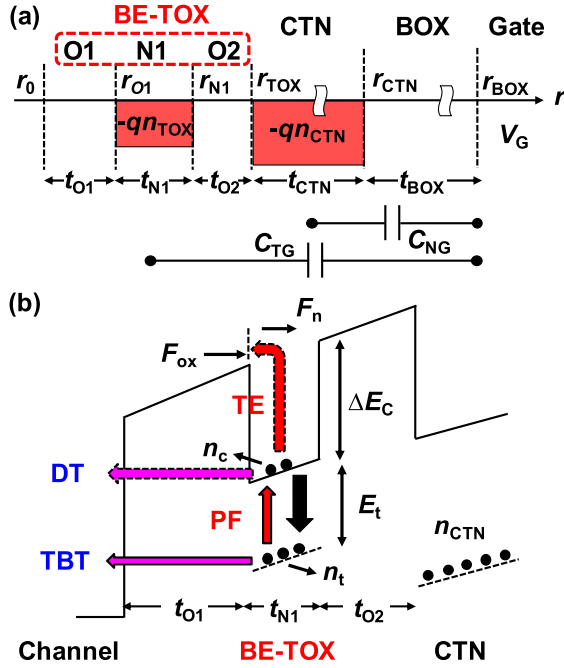


Fig. 1. (a) Charge distribution and the capacitive couplings (C_{NG} and C_{TG}) in BE-TOX/CTN/BOX layers in cylindrical coordinates (r). (b) Magnified energy band diagram and the directions of the electron dynamics in the detrapping mechanism.

TABLE I
NOTATIONS FOR MODEL PARAMETERS USED IN THIS ARTICLE

Symbol	Physical meaning	Value
r_0	Channel radius	25 nm [19]
$t_{O1}/t_{N1}/t_{O2}$	Bandgap engineered tunneling oxide (BE-TOX) thickness	2/2/2 nm
t_{CTN}	Charge-trap nitride (CTN) thickness	5 nm [19]
t_{BOX}	Blocking oxide (BOX) thickness	10 nm [19]
ϵ_{ox}	The dielectric constant of silicon dioxide	$3.9\epsilon_0$ [20]
ϵ_n	The dielectric constant of silicon nitride	$7.5\epsilon_0$ [20]
m_{ox}	The effective mass of silicon dioxide	$0.5m_0$ [20]
m_n	The effective mass of silicon nitride	$0.4m_0$ [20]
N_t	Total trap density of BE-TOX trap	$6 \times 10^{17} \text{ cm}^{-3}$
N_{CTN}	Total trap density of CTN trap	$5 \times 10^{19} \text{ cm}^{-3}$ [20]
σ_n	Electron capture cross section of BE-TOX trap	$6 \times 10^{-15} \text{ cm}^2$ [20]
ΔE_C	Conduction band offset between oxide and nitride	0.8 eV [20]
E_t	BE-TOX trap energy level	0.6 eV [10]

II. COMPACT MODELING

A. Compact Model Description

Fig. 1(a) shows the charge distribution in BE-TOX/CTN/BOX layers in cylindrical coordinates (r). By separating the BE-TOX layer into three layers (O1/N1/O2) among which the uniform charge distribution ($-qn_{TOX}$) is defined in the nitride layer (N1), we solved 1-D Poisson equations for entire five layers by the similar ways (but with more boundary conditions) in [15] and [18]. From the solutions, we obtained the equations for the electric field (E -field) which depend on the trapped charge densities and the geometrical parameters. Table I summarizes the model parameters relevant to the

device geometry and the trap physics. To consider the barrier lowering due to the E -field, the E -field equations in the O1 and the N1 regions are respectively obtained as

$$F_{O1}(r) = -\frac{K}{\epsilon_{ox}r} \quad (1)$$

$$F_{N1}(r) = -\frac{K - qn_{TOX}(r_{O1}^2 - r^2)/2}{\epsilon_n r} \quad (2)$$

where the parameter K and its internal constant α are given by

$$K = \frac{1}{\alpha} \left(V_G - \frac{qn_{TOX}VOL_{N1}}{C_{TG}} - \frac{qn_{CTN}VOL_{CTN}}{C_{NG}} \right) \quad (3)$$

$$\alpha = \frac{1}{\epsilon_{ox}} \ln \frac{r_{O1}}{r_0} + \frac{1}{\epsilon_n} \ln \frac{r_{N1}}{r_{O1}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{TOX}}{r_{N1}} + \frac{1}{\epsilon_n} \ln \frac{r_{CTN}}{r_{TOX}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{BOX}}{r_{CTN}} \quad (4)$$

In the above equations, V_G is the gate bias, and n_{TOX} and n_{CTN} are the electron densities (units: cm^{-3}) in the N1 and in the CTN layers, respectively. n_{TOX} is the sum of the trapped electron density (n_t) and the conduction electron density (n_c). VOL_{N1} and VOL_{CTN} are the volumes (units: cm^3) of the N1 and CTN layers, respectively. Finally, C_{TG} and C_{NG} are the cylindrical capacitances (units: F) between the N1 layer and the gate and between the CTN layer and the gate, respectively. Fig. 1(b) shows the energy band diagram of the BE-TOX while the detrapping occurs. From the acceptor-like shallow trap of the N1 layer, the trapped electrons are released via TBT toward the channel or via Poole-Frenkel (PF) emission toward the conduction band. Each emission rate via those components is modeled as [11], [13]

$$e_{TBT} = v_{TBT} \exp(-\alpha_{ox,T} t_{O1}) \cdot \frac{1}{t_{N1}} \int_0^{t_{N1}} \exp(-\alpha_{n,T} x) dx \\ = v_{TBT} \exp(-\alpha_{ox,T} t_{O1}) \cdot \frac{1 - \exp(-\alpha_{n,T} t_{N1})}{\alpha_{n,T} t_{N1}} \quad (5)$$

$$e_{PF} = 2\sigma_n \sqrt{\frac{3k_B}{m_n}} \left(\frac{2\pi m_n k_B}{h} \right)^{3/2} T^2 \cdot \exp\left(\frac{\beta_n \sqrt{F_n} - E_t}{k_B T} \right) \quad (6)$$

where v_{TBT} is TBT attempt-to-escape frequency and $\alpha_{ox,T}$ and $\alpha_{n,T}$ in (5) are related to the tunneling barriers of the trapped electrons through the O1 and N1 layers, respectively. They are calculated as $4\sqrt{2m_{ox}}(E_t + \Delta E_C - \beta_{ox}\sqrt{F_{ox}})^{3/2} \alpha'_{TBT}/(3\hbar q)$ and $2\sqrt{2m_n E_t}/\hbar$, respectively. α'_{TBT} is used as a fitting parameter and calibrates the e_{TBT} dependence on the tunneling distance (t_{O1}) as modeled in the gate oxide tunneling current of Berkeley short-channel IGFET model-common multi-gate (BSIM-CMG) model [21]. Since the distribution of the trapped electrons in the N1 layer is not always uniform in practice and is variable during the retention time, the actual tunneling distance can be different from t_{O1} . Therefore, α'_{TBT} can be seen as a fitting parameter which also corrects those variations. In (5), we calculated the average value of e_{TBT} to effectively consider the decaying tunneling probability along the thin N1 layer. Also, contrary to the CTN charge loss model in [13], we considered the energy barrier lowerings of the O1 and of the N1 layers due to the E -fields (F_{ox} and F_n) induced by the CTN charge as $\beta_{ox} = \sqrt{q^3/(\pi\epsilon_{ox})}$ and $\beta_n = \sqrt{q^3/(\pi\epsilon_n)}$,

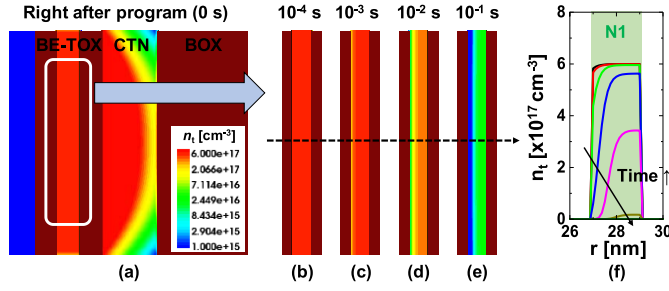


Fig. 2. Trapped electrons contours (a) right after the program (0 s), at (b) 10^{-4} s, (c) 10^{-3} s, (d) 10^{-2} s, and (e) 10^{-1} s. (f) Radial distributions of the trapped electrons over the retention time. They are obtained at the middle part of the channel.

respectively. The contributing E -fields F_{ox} and F_n are the average values of $F_{O1}(r)$ and $F_{N1}(r)$, respectively. Then, the electrons elevated to the conduction band can subsequently escape toward the channel by DT or by TE. Each emission rate via the components is modeled as [11]

$$e_{DT} = v_{DT} \cdot \exp(-\alpha_{ox,C} t_{O1}) \quad (7)$$

$$e_{TE} = v_{TE} \cdot \exp\left(\frac{\beta_{ox}\sqrt{F_{ox}} - \Delta E_C}{k_B T}\right) \quad (8)$$

where v_{DT} and v_{TE} are DT attempt-to-escape frequency and TE attempt-to-escape frequency, respectively. $\alpha_{ox,C}$ in (7) is related to the tunneling barrier of the conduction electrons through the O1 layer and is expressed as $4\sqrt{2m_{ox}}(\Delta E_C - \beta_{ox}\sqrt{F_{ox}})^{3/2} \alpha'_{DT}/(3\hbar q)$. Similar to α'_{TBT} in TBT, α'_{DT} is also used as a fitting parameter and corrects the variations of the electron distribution. Simultaneously, the electrons on the conduction band can be captured again into the trap by the capture rate of $c_n = v_{th}\sigma_n(N_t - n_t)$ [11] where the thermal velocity (v_{th}) is calculated as $\sqrt{3k_B T/m_n}$. Then, the rates of n_t and n_c and the consequent ΔV_{th} due to the detrapping are calculated as follows:

$$\frac{dn_t}{dt} = c_n \cdot n_c - (e_{TBT} + e_{PF}) \cdot n_t \quad (9)$$

$$\frac{dn_c}{dt} = e_{PF} \cdot n_t - (e_{DT} + e_{TE} + c_n) \cdot n_c \quad (10)$$

$$\Delta V_{th} = -q \Delta n_{TOX} \text{VOL}_{N1} / C_{TG}. \quad (11)$$

Typically, the electrons stored in the CTN layer are known to escape through the trap-assisted tunneling (TAT) across the TOX traps [22], [23]. However, such emission in 3-D NAND flash memories with BE-TOX is reported to occur after much longer retention time [7], and the detrapping mechanism generally focuses on the electrons present in the TOX, thus TAT is not discussed in this study.

B. Model Calibration

From the abovementioned model, we calibrated our model to the results from the 3-D TCAD simulation. The TCAD simulation is a suitable method for separating the complex components included in the detrapping mechanism without the interferences of other charge loss mechanisms. Fig. 2 shows the cross section of the BE-TOX layer with the transition of the

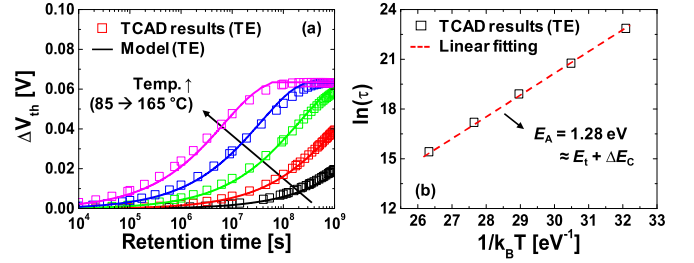


Fig. 3. (a) Calibration of the model to the simulation results for TE ($t_{O1} = 2$ nm). The extracted value of v_{TE} is $3.0 \times 10^5 \text{ s}^{-1}$. The oxides are located at both edges of BE-TOX to prevent the lateral spreading of conduction electrons. (b) Natural logarithms of τ and a linearly fitted line for different temperatures for TE.

trapped electrons contours over the retention time. These contours are extracted from the TCAD simulation which includes the necessary physical models such as PF, eThermionic, DT, and eBarrierTunneling with nonlocal model. The values of other parameters in the simulation were kept the same as those of the model. Fig. 2(a) shows the trapped electrons contour right after the program operation and the BE-TOX traps are almost full. Over the retention time, the trapped electrons in the BE-TOX are emitted as shown in Fig. 2(b)–(e). The radial distributions of the trapped electrons over the retention time are obtained at the middle part of the channel and shown in Fig. 2(f). To accurately consider each physics in the BE-TOX trap, we calibrated our model in the order of TE, DT, and TBT. Before the detrapping, we increased the V_{th} by 1 V by the program simulation with nonlocal tunneling model [16], and then obtained the initial parameter values of n_t and n_{CTN} .

First, to calibrate TE, we activated the physics of PF and eThermionic on the TCAD simulation and set e_{DT} and e_{TBT} to 0 in the model. The symbols in Fig. 3(a) show the simulation results of the ΔV_{th} due to the electron detrapping via TE for different temperatures (85 °C–165 °C). In these simulations, we located oxides with high energy barrier at both edges of the BE-TOX of the cell so that the ΔV_{th} occurs only by TE, not by the lateral spreading of conduction electrons. As shown, the time constants (τ s) are significantly larger than those of the common short-term retention. This is because the emitted electrons from the trap should exceed ΔE_C again which is larger than E_t . For the model implementation, we extracted the value of v_{TE} in (8) by fitting the model, and the value is $3.0 \times 10^5 \text{ s}^{-1}$. The model results are shown as the lines. Fig. 3(b) shows the natural logarithms of τ for different temperatures. The slope of the linearly fitted $\ln(\tau)$ s (i.e., activation energy) is extracted to be 1.28 eV and this value is comparable to $E_t + \Delta E_C$. Due to its large time constants and high activation energy, TE component has almost negligible effect in the short-term retention.

When applying the bandgap engineering technology, the position of the N1 layer in the BE-TOX can shift depending on the process condition. Before the analysis of detrappings with various t_{O1} , we extracted the initial ΔV_{th} after program operation for different t_{O1} . Fig. 4(a) and (b) shows the energy band diagrams (black lines) during the program operations ($V_G = 16$ V) and the distributions of the trapped

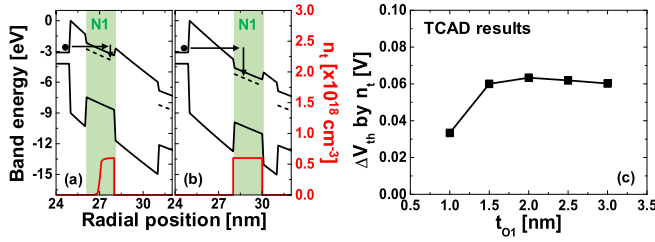


Fig. 4. Energy band diagrams (black lines) during the program operations ($V_G = 16$ V) and the distributions of the trapped electrons in the BE-TOX (red lines) after the operations for (a) $t_{O1} = 1$ nm and (b) $t_{O1} = 3$ nm. (c) ΔV_{th} by the trapped electrons in the BE-TOX after program operation for different t_{O1} .

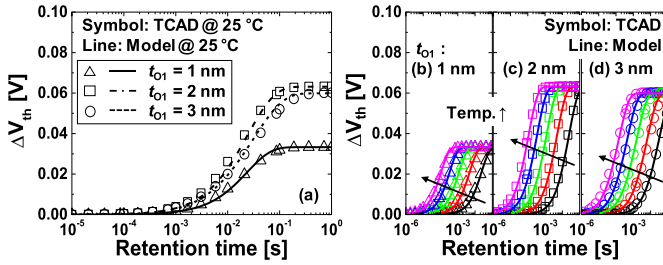


Fig. 5. (a) Model calibration to the simulation results with TE and DT at 25 °C for different t_{O1} . The fitting parameters are $\nu_{DT} = 4.0 \times 10^{11} \text{ s}^{-1}$ and $\alpha'_{DT} = 0.231 \text{ V}^{-1}$. The results of the other temperatures are included for (b) $t_{O1} = 1$ nm, (c) $t_{O1} = 2$ nm, and (d) $t_{O1} = 3$ nm.

electrons in the BE-TOX (red lines) after programs for $t_{O1} = 1$ nm and $t_{O1} = 3$ nm, respectively. When t_{O1} is thin as 1 nm, the tunneling barrier shape corresponds to the modified Fowler–Nordheim (FN) region as shown in Fig. 4(a). In this case, the N1 is divided into two regions: one is a region where the electrons are trapped and the other is a region where the electrons are tunneling through. Therefore, the electrons are mostly trapped at the outer edge of the N1 layer. On the other hand, when t_{O1} is thick as 3 nm, the tunneling barrier shape becomes triangular (FN region) and the trapped electrons are extended to the inner side of the N1 layer as shown in Fig. 4(b). Fig. 4(c) shows ΔV_{th} by the trapped electrons in the BE-TOX after program operation for different t_{O1} . From $t_{O1} = 1$ nm to $t_{O1} = 2$ nm, ΔV_{th} increases due to the increased amount of electrons in the BE-TOX. After $t_{O1} = 2$ nm, ΔV_{th} slightly decreases since the increase in C_{TG} is dominant rather than the change in the amount of electrons.

Then, we included the physics of DT on the TCAD simulation and set e_{TBT} to 0 in the model. The extracted detrapping characteristics at 25 °C for different t_{O1} are shown in the symbols in Fig. 5(a). By modulating the parameters ν_{DT} and α'_{DT} , we fitted the model results (lines) to the symbols. ν_{DT} determines the overall e_{DT} irrespective of t_{O1} , and α'_{DT} determines the τ difference depending on t_{O1} . The extracted values of the parameters are $\nu_{DT} = 4.0 \times 10^{11} \text{ s}^{-1}$ and $\alpha'_{DT} = 0.231 \text{ V}^{-1}$. Fig. 5(b)–(d) shows the detrapping characteristics via TE and DT for different temperatures and t_{O1} . Since the faster detrapping at the increased temperature is caused by the faster PF (not by the increased e_{DT}), the model results are well matched to TCAD results at the other temperatures without modulating additional parameters.

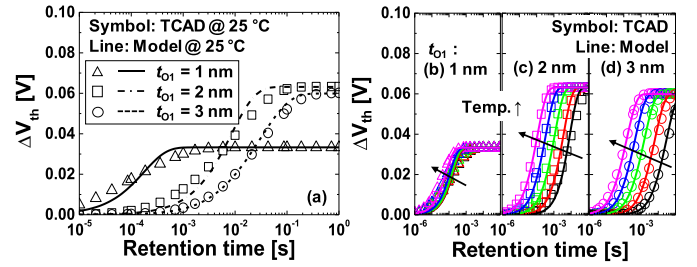


Fig. 6. (a) Model calibration to the simulation results with TE, DT, and TBT at 25 °C for different t_{O1} . The fitting parameters are $\nu_{TBT} = 4.2 \times 10^6 \text{ s}^{-1}$ and $\alpha'_{DT} = 0.537 \text{ V}^{-1}$. The results of the other temperatures are included for (b) $t_{O1} = 1$ nm, (c) $t_{O1} = 2$ nm, and (d) $t_{O1} = 3$ nm.

Finally, we included the physics of eBarrierTunneling with nonlocal model on the TCAD simulation to activate TBT and extracted detrapping characteristics at 25 °C for different t_{O1} as shown in the symbols in Fig. 6(a). Similar to DT modeling in Fig. 5, we fitted the model results (lines) to the symbols by modulating the parameters ν_{TBT} and α'_{TBT} . The extracted values of the parameters are $\nu_{TBT} = 4.2 \times 10^6 \text{ s}^{-1}$ and $\alpha'_{TBT} = 0.537 \text{ V}^{-1}$. Compared with the DT results in Fig. 5(a), the τ decreased significantly for the case of $t_{O1} = 1$ nm, decreased slightly for the case of $t_{O1} = 2$ nm, and there is little difference in τ for the case of $t_{O1} = 3$ nm. This indicates that e_{TBT} is more sensitive to t_{O1} than e_{DT} . Fig. 6(b)–(d) shows the detrapping characteristics via TE, DT, and TBT for different temperatures and t_{O1} , and the model results are well matched to the TCAD results at the other temperatures.

III. RESULTS AND DISCUSSION

From the detrapping model and the determined set of the model parameters in the previous section, we analyzed the rates of each component and the influences of each emission path for various temperatures and N1 layer positions.

A. Dependence on Temperature and N1 Layer Position

Fig. 7(a)–(c) shows each rate of the detailed components (e_{TBT} , e_{PF} , c_n , e_{DT} , and e_{TE}) in log scale for the case of $t_{O1} = 2$ nm at 25 °C, 65 °C, and 105 °C, respectively. For increasing temperature, the rates of PF and TE increase while the rates of TBT and DT are almost the same. e_{PF} and e_{TE} increase by approximately one order as the temperature increases by 40 °C. In order to intuitively compare the amounts of charge loss via each path, we calculated the carrier density rates via each path (units: $\text{cm}^{-3}\text{s}^{-1}$) by the product of the rate coefficient and the carrier density. The density rates via TBT, DT, and TE are denoted as D_{TBT} , D_{DT} , and D_{TE} , respectively. Each of them is calculated as

$$D_{TBT} = e_{TBT} \cdot n_t \quad (12)$$

$$D_{DT} = e_{DT} \cdot n_c \quad (13)$$

$$D_{TE} = e_{TE} \cdot n_c. \quad (14)$$

The sum of the three density rates ($D_{TBT} + D_{DT} + D_{TE}$) is equal to the rate of the total charge loss from the BE-TOX ($-dn_{TOX}/dt$). The calculated density rates are shown in log

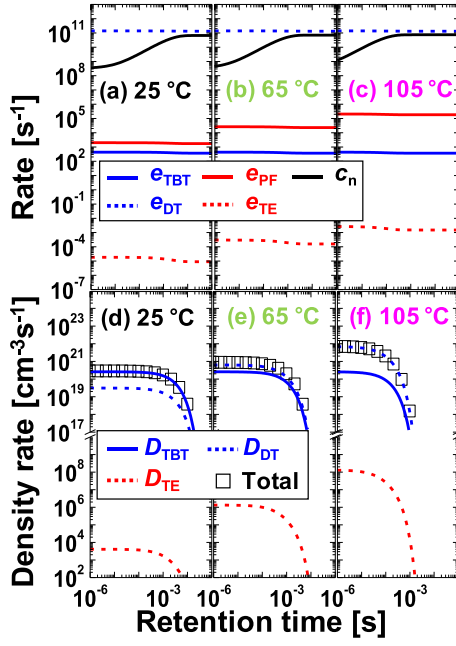


Fig. 7. Extracted rates of the detailed components (e_{TBT} , e_{PF} , c_n , e_{DT} , and e_{TE}) at (a) 25 °C, (b) 65 °C, and (c) 105 °C for the case of $t_{O1} = 2$ nm. The calculated density rates via each path (D_{TBT} , D_{DT} , and D_{TE}) at (d) 25 °C, (e) 65 °C, and (f) 105 °C.

scale in Fig. 7(d)–(f) for different temperatures of 25 °C, 65 °C, and 105 °C, respectively. The results show that the major path of the charge loss is TBT at 25 °C, and the influence of DT increases as the temperature increases. On the other hand, TE component has negligible influences within the temperature range in this study.

Fig. 8(a) and (b) shows each rate of the detailed components for $t_{O1} = 1$ nm at 25 °C and 105 °C, respectively. Since the tunneling length is very short as 1 nm, e_{TBT} is higher than e_{PF} at 25 °C. When the temperature increases to 105 °C, e_{PF} and e_{TE} increase by approximately two orders. Using the rates in Fig. 8(a) and (b), the density rates are also calculated and shown in Fig. 8(c) and (d) for the case of $t_{O1} = 1$ nm. Since e_{TBT} is higher than e_{PF} at 25 °C, TBT is the major path of the charge loss at 25 °C. Even though e_{PF} is higher than e_{TBT} at 105 °C, the actual amount of the electrons which escape from the conduction band is comparable to the amount of the released electrons via TBT, since a large portion of the elevated electrons is trapped again when $t_{O1} = 1$ nm. When the position of the N1 shifts toward the CTN as $t_{O1} = 3$ nm, the difference between e_{TBT} and e_{PF} significantly increases as shown in Fig. 8(e) and (f). Therefore, TBT no longer affects the total charge loss and DT from the conduction band becomes the major path on a wide range of temperatures [Fig. 8(g) and (h)]. Within the t_{O1} range in this study, TE does not affect the total charge loss due to its significantly low emission rate.

B. Curve Shape and Activation Energy

For further quantitative analysis on the curve shapes and the activation energies for various t_{O1} , we adopted a stretched

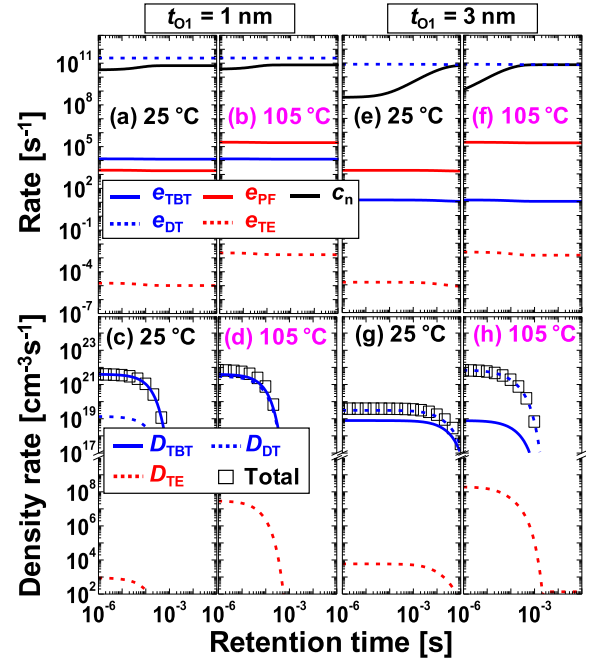


Fig. 8. (a) Extracted rates of the detailed components (e_{TBT} , e_{PF} , c_n , e_{DT} , and e_{TE}) for (a) $t_{O1} = 1$ nm at 25 °C, (b) $t_{O1} = 1$ nm at 105 °C, (e) $t_{O1} = 3$ nm at 25 °C, and (f) $t_{O1} = 3$ nm at 105 °C. The calculated density rates via each path (D_{TBT} , D_{DT} , and D_{TE}) are shown in (c), (d), (g), and (h) in the same order.

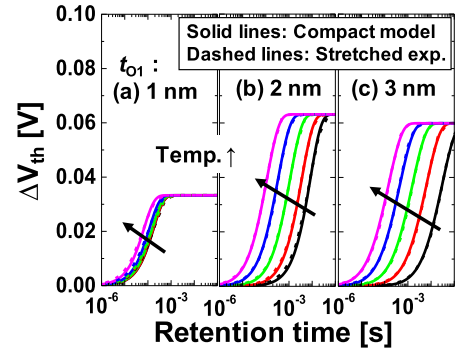


Fig. 9. Model results (solid lines) of ΔV_{th} by the detrapping with shifting N1 layer to (a) $t_{O1} = 1$ nm, (b) $t_{O1} = 2$ nm, and (c) $t_{O1} = 3$ nm. Each curve is fitted with the stretched exponential functions (dashed lines) to extract $\tau_{Detrapping}$ and $\beta_{Detrapping}$.

exponential function of the detrapping mechanism as [4]

$$\Delta V_{th}(t) = \Delta V_{th, Detrapping} \cdot \left[1 - \exp\left(-\left(\frac{t}{\tau_{Detrapping}}\right)^{\beta_{Detrapping}}\right) \right] \quad (15)$$

where $\Delta V_{th, Detrapping}$ is the total V_{th} loss by the detrapping [corresponding to Fig. 4(c)], $\beta_{Detrapping}$ determines the curve shape (slope) of the ΔV_{th} at the retention time t , and $\tau_{Detrapping}$ is the time constant of each ΔV_{th} curve by the detrapping. Then we extracted the values of $\tau_{Detrapping}$ and $\beta_{Detrapping}$ for each modeling result (solid line) in Fig. 9 by overlapping the fitted stretched exponential functions as the dashed lines. We used

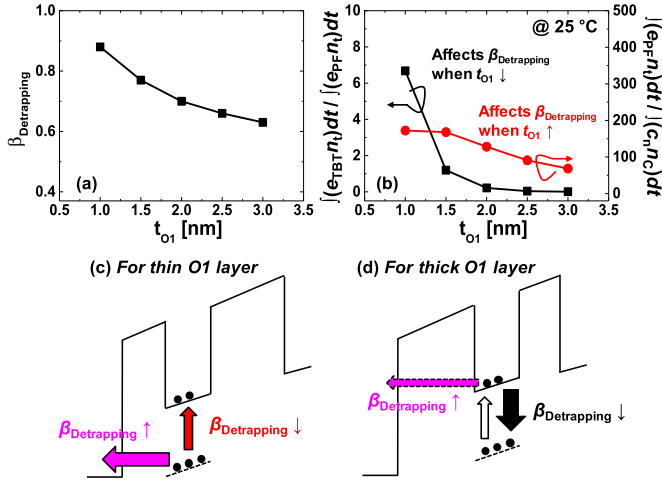


Fig. 10. (a) Extracted $\beta_{\text{Detrapping}}$ from the fitted stretched exponential functions for different t_{O1} . (b) Trend of $\beta_{\text{Detrapping}}$ is comparable to the trends of $\int (e_{\text{TBT}} n_i) dt / \int (e_{\text{PF}} n_i) dt$ for thin O1 and $\int (e_{\text{PF}} n_i) dt / \int (c_n n_c) dt$ for thick O1 (at 25 °C). The band diagrams and the detailed components which affect $\beta_{\text{Detrapping}}$ when the O1 layers are (c) thin and (d) thick.

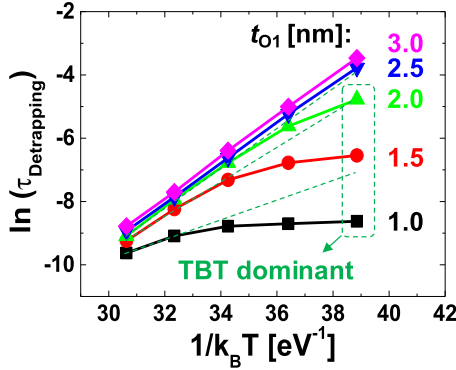


Fig. 11. Extracted $\ln(\tau_{\text{Detrapping}})$ from the fitted stretched exponential functions for different t_{O1} .

a unified value of $\beta_{\text{Detrapping}}$ (<1) for every temperature in the same t_{O1} by following the limiting conditions in [6].

Using the carrier density rates, we analyzed the retention characteristics (in terms of $\tau_{\text{Detrapping}}$ and $\beta_{\text{Detrapping}}$) extracted in Fig. 9 for varying t_{O1} . Fig. 10(a) shows the extracted $\beta_{\text{Detrapping}}$ as a function of t_{O1} . $\beta_{\text{Detrapping}}$ increases as t_{O1} decreases, which indicates that the ΔV_{th} curve shape becomes sharper as the N1 layer shifts toward the channel. First, we compared this trend of $\beta_{\text{Detrapping}}$ with the trend of the ratio between the integrated density rates via TBT and PF $[\int (e_{\text{TBT}} n_i) dt / \int (e_{\text{PF}} n_i) dt]$ at 25 °C as the black squares in Fig. 10(b). Contrary to TBT which is a 1-step emission process, PF is followed by the recapture component. This factor reduces the efficiency of the charge loss and finally leads to the decrease in $\beta_{\text{Detrapping}}$. Therefore, when the O1 layer is sufficiently thin so that TBT is dominant, the value of $\beta_{\text{Detrapping}}$ is close to 1 and gradually decreases as PF is involved as described in Fig. 10(c). Then, we compared the trend of $\beta_{\text{Detrapping}}$ with the trend of the ratio between the integrated density rates via PF and recapture $[\int (e_{\text{PF}} n_i) dt / \int (c_n n_c) dt]$ at 25 °C as the red circles in Fig. 10(b). TBT component is

almost negligible and PF component becomes more dominant as the O1 layer becomes thicker. Since DT component decreases for increasing t_{O1} , the portion of the recaptured electrons increases. For this reason, the value of $\beta_{\text{Detrapping}}$ decreases for thick O1 layer as described in Fig. 10(d).

Fig. 11 shows the natural logarithms of $\tau_{\text{Detrapping}}$ as functions of $1/k_B T$ for various t_{O1} . The slopes of $\ln(\tau_{\text{Detrapping}})$ s (i.e., activation energies) for large t_{O1} are almost constant while those slopes for small t_{O1} gradually decrease as the temperature decreases. The reason is that TBT does not depend on the temperature and TBT contributes more as t_{O1} decreases or the temperature decreases. Since the portion of D_{TBT} among the total charge loss is the lowest when $t_{\text{O1}} = 3$ nm, the activation energy for the case of $t_{\text{O1}} = 3$ nm is the largest (~ 0.56 eV) among every t_{O1} .

IV. CONCLUSION

We newly proposed a fast detrapping model in 3-D NAND flash memories with the possible charge loss paths from the shallow trap of the BE-TOX. While the detrapping from the TOX has been generally considered and analyzed as a single mechanism among the various charge loss mechanisms, we separated this mechanism into the detailed paths and investigated the retention characteristics dependence on the various physical conditions for the first time. The TBT component is highly influenced by the position of the N1 in the BE-TOX (by about three orders for 2 nm change in t_{O1}), and the TE component is found to be almost negligible due to its high effective barrier height ($E_t + \Delta E_c$). Especially, the thicker inner O1 layer causes a slower V_{th} loss, but has a larger activation energy and a larger ΔV_{th} . Therefore, our compact model not only provides an efficient tool for the evaluation of the electron detrapping from the BE-TOX but also provides physical insights of this mechanism in terms of the more detailed charge loss paths.

REFERENCES

- [1] H. Kim, S.-J. Ahn, Y. G. Shin, K. Lee, and E. Jung, "Evolution of NAND flash memory: From 2D to 3D as a storage market leader," in *Proc. IEEE Int. Memory Workshop*, Monterey, CA, USA, May 2017, pp. 1–4, doi: [10.1109/IMW.2017.7939081](https://doi.org/10.1109/IMW.2017.7939081).
- [2] E.-S. Choi and S.-K. Park, "Device considerations for high density and highly reliable 3D NAND flash cell in near future," in *IEDM Tech. Dig.*, Dec. 2012, pp. 9.4.1–9.4.4, doi: [10.1109/IEDM.2012.6479011](https://doi.org/10.1109/IEDM.2012.6479011).
- [3] A. Goda, "3-D NAND technology achievements and future scaling perspectives," *IEEE Trans. Electron Devices*, vol. 67, no. 4, pp. 1373–1381, Apr. 2020, doi: [10.1109/TED.2020.2968079](https://doi.org/10.1109/TED.2020.2968079).
- [4] K. Lee, M. Kang, Y. Hwang, and H. Shin, "Accurate lifetime estimation of Sub-20-nm NAND flash memory," *IEEE Trans. Electron Devices*, vol. 63, no. 2, pp. 659–667, Feb. 2016, doi: [10.1109/TED.2015.2509004](https://doi.org/10.1109/TED.2015.2509004).
- [5] D. Oh *et al.*, "TCAD Simulation of data retention characteristics of charge trap device for 3-D NAND flash memory," in *Proc. IMW*, 2015, pp. 1–4, doi: [10.1109/IMW.2015.7150306](https://doi.org/10.1109/IMW.2015.7150306).
- [6] C. Woo *et al.*, "Modeling of charge loss mechanisms during the short term retention operation in 3-D NAND flash memories," in *VLSI Symp. Tech. Dig.*, 2019, pp. 214–215, doi: [10.23919/VLSIT.2019.8776579](https://doi.org/10.23919/VLSIT.2019.8776579).
- [7] C. Woo *et al.*, "Modeling of charge failure mechanisms during the short term retention depending on program/erase cycle counts in 3-D NAND flash memories," in *Proc. IEEE IRPS*, Apr. 2020, pp. 1–6, doi: [10.1109/IRPS45951.2020.9129306](https://doi.org/10.1109/IRPS45951.2020.9129306).
- [8] H.-T. Lue *et al.*, "BE-SONOS: A bandgap engineered SONOS with excellent performance and reliability," in *IEDM Tech. Dig.*, Oct. 2005, pp. 547–550, doi: [10.1109/IEDM.2005.1609404](https://doi.org/10.1109/IEDM.2005.1609404).

- [9] N. Choi, H.-J. Kang, J.-H. Bae, B.-G. Park, and J.-H. Lee, "Effect of nitrogen content in tunneling dielectric on cell properties of 3-D NAND flash cells," *IEEE Electron Device Lett.*, vol. 40, no. 5, pp. 702–705, May 2019, doi: [10.1109/LED.2019.2905299](https://doi.org/10.1109/LED.2019.2905299).
- [10] J. Wu, D. Han, W. Yang, S. Chen, X. Jiang, and J. Chen, "Comprehensive investigations on charge diffusion physics in SiN-based 3D NAND flash memory through systematical Ab Initio calculations," in *IEDM Tech. Dig.*, Dec. 2007, pp. 4–5, doi: [10.1109/IEDM.2017.8268327](https://doi.org/10.1109/IEDM.2017.8268327).
- [11] S.-H. Gu, C.-W. Hsu, T. Wang, W.-P. Lu, Y.-H.-J. Ku, and C.-Y. Lu, "Numerical simulation of bottom oxide thickness effect on charge retention in SONOS flash memory cells," *IEEE Trans. Electron Devices*, vol. 54, no. 1, pp. 90–97, Jan. 2007, doi: [10.1109/TED.2006.887219](https://doi.org/10.1109/TED.2006.887219).
- [12] K. Bernert, C. Oestreich, J. Bollmann, and T. Mikolajick, "The influence of bottom oxide thickness on the extraction of the trap energy distribution in SONOS (silicon-oxide-nitride-oxide-silicon) structures," *Appl. Phys. A, Solids Surf.*, vol. 100, no. 1, pp. 249–255, Apr. 2010, doi: [10.1007/s00339-010-5694-0](https://doi.org/10.1007/s00339-010-5694-0).
- [13] Y. Yang and M. H. White, "Charge retention of scaled SONOS nonvolatile memory devices at elevated temperatures," *Solid-State Electron.*, vol. 44, no. 6, pp. 949–958, Jan. 2000, doi: [10.1016/S0038-1101\(00\)00012-5](https://doi.org/10.1016/S0038-1101(00)00012-5).
- [14] A. Arreghini, N. Akil, F. Driussi, D. Esseni, L. Selmi, and M. J. van Duuren, "Long term charge retention dynamics of SONOS cells," *Solid-State Electron.*, vol. 52, no. 9, pp. 1460–1466, Sep. 2008, doi: [10.1016/j.sse.2008.04.016](https://doi.org/10.1016/j.sse.2008.04.016).
- [15] M. Kim, S. Kim, and H. Shin, "A compact model for ISPP of 3-D charge-trap NAND flash memories," *IEEE Trans. Electron Devices*, vol. 67, no. 8, pp. 3095–3101, Aug. 2020, doi: [10.1109/TED.2020.3000448](https://doi.org/10.1109/TED.2020.3000448).
- [16] *TCAD Sentaurus Device User Guide, Version 2015*, Synopsys, Mountain View, CA, USA, Jun. 2015.
- [17] K. Lee *et al.*, "Activation energies (E_a) of failure mechanisms in advanced NAND flash cells for different generation and cycling," *IEEE Trans. Electron Devices*, vol. 60, no. 3, pp. 1099–1107, Mar. 2013, doi: [10.1109/TED.2013.2241065](https://doi.org/10.1109/TED.2013.2241065).
- [18] S. M. Amoroso, C. Monzio Compagnoni, A. Mauri, A. Maconi, A. S. Spinelli, and A. L. Lacaita, "Semi-analytical model for the transient operation of gate-all-around charge-trap memories," *IEEE Trans. Electron Devices*, vol. 58, no. 9, pp. 3116–3123, Sep. 2011, doi: [10.1109/TED.2011.2159010](https://doi.org/10.1109/TED.2011.2159010).
- [19] M. Kim and H. Shin, "Investigation and SPICE compact model of spacer region for static characteristics of 3-D NAND flash memories," *IEEE Trans. Electron Devices*, vol. 67, no. 10, pp. 4158–4165, Oct. 2020, doi: [10.1109/TED.2020.3014066](https://doi.org/10.1109/TED.2020.3014066).
- [20] A. Padovani *et al.*, "A comprehensive understanding of the erase of TANOS memories through charge separation experiments and simulations," *IEEE Trans. Electron Devices*, vol. 58, no. 9, pp. 3147–3155, Sep. 2011, doi: [10.1109/TED.2011.2159722](https://doi.org/10.1109/TED.2011.2159722).
- [21] (Jan. 2016). *BSIM-CMG 110.0.0 Technical Manual*. [Online]. Available: <https://www.bsim.berkeley.edu/models/bsimcmg/>
- [22] A. Modelli, F. Gilardoni, D. Ielmini, and A. S. Spinelli, "A new conduction mechanism for the anomalous cells in thin oxide Flash EEPROMs," in *Proc. IEEE IRPS*, Apr. 2001, pp. 1–8, doi: [10.1109/RELPHY.2001.922883](https://doi.org/10.1109/RELPHY.2001.922883).
- [23] D. Ielmini, A. S. Spinelli, A. L. Lacaita, and A. Modelli, "A statistical model for SILC in flash memories," *IEEE Trans. Electron Devices*, vol. 49, no. 11, pp. 1955–1961, Nov. 2002, doi: [10.1109/TED.2002.804730](https://doi.org/10.1109/TED.2002.804730).